



EXL Data Engineer Interview Question

Data Engineer

Round: 1

Interview Questions with Answers

1. How do you handle Slowly Changing Dimensions in ETL?

Answer:

Genpact commonly uses SCD Type 2.

Steps:

1. Compare source vs existing dimension.
2. If any attribute changes:
 - o Close old record by setting End_Date.
 - o Insert new record with updated values.
 - o Mark Is_Current = True.
3. Maintain Effective_Start_Date, Effective_End_Date.

Used in: Customer profile history, employee data, product catalog.

2. SQL to delete duplicates and keep only the latest row

```
WITH cte AS (  
    SELECT *,  
           ROW_NUMBER() OVER(PARTITION BY customer_id ORDER BY updated_at  
DESC) AS rn  
    FROM customers  
)  
DELETE FROM cte WHERE rn > 1;
```

3. Explain design of a daily incremental data pipeline.

Answer:

1. Identify incremental rows using timestamp, last modified column, or high watermark.

2. Load only new/changed records.
3. Use upsert/merge logic on the target.
4. Track audit details:
 - o batch_id
 - o success_count
 - o failure_count
5. Add error-handling and retries.

4. How do you optimize a slow SQL query?

Answer:

- Review execution plan.
- Add missing indexes.
- Avoid functions on indexed columns.
- Use EXISTS instead of IN.
- Use partitioning for large fact tables.
- Reduce subqueries, rewrite joins.

5. Difference between INNER JOIN and LEFT JOIN

INNER JOIN: Returns only matching records.

LEFT JOIN: Returns all records from left side, and matching from right.

6. Real Scenario: Your table has 500M rows and the query is too slow. What do you do?

Answer:

- Partition table on date or category.
- Use clustered columnstore indexes (SQL Server).
- Move old data to archive tables.
- Avoid SELECT * and remove unnecessary joins.
- Use CTAS (Create Table As Select) for optimization in Azure Synapse.

7. SQL to find the second highest salary

```
SELECT MAX(salary)
FROM employees
WHERE salary < (SELECT MAX(salary) FROM employees);
```

8. What is a Data Lake and how is it used in Genpact?

Answer:

A Data Lake stores raw/structured/semi-structured data.

Genpact typically uses **Azure Data Lake (ADLS)** with:



- Bronze: Raw data
- Silver: Cleaned, validated
- Gold: Curated, business-ready

Used in Customer 360, Risk Analytics, Claims Processing.

9. Python code to read a CSV and remove nulls

```
import pandas as pd

df = pd.read_csv("input.csv")
df = df.dropna()
df.to_csv("cleaned.csv", index=False)
```

10. Difference between OLTP and OLAP

Feature	OLTP	OLAP
Purpose	Transaction processing	Analytics
Data	Current	Historical
Operations	Insert/update	Aggregation & reporting
Example	Banking app	Data Warehouse

11. A pipeline failed in production. What is your approach?

1. Check logs (pipeline, activity, execution logs).
2. Locate failure point.
3. Validate input file and schema.
4. Re-run failed batches.
5. Fix logic issue or data quality issue.
6. Add monitoring → alerts, retries, fallbacks.

12. How do you perform incremental loads in ADF?

Answer:

- Maintain a watermark table.
- Lookup last loaded timestamp.
- Filter source: WHERE last_updated > watermark.
- Sink: Use MERGE/Upsert.
- Update watermark after successful load.

